

Lab 1 SQL DML Practical Problems (by partners)

General Instructions

In separate files named `hospital1.sql`, `hospital2.sql`, ..., `bookstore1.sql`, `bookstore2.sql`, ..., do the following practical problems. Place each problem's SQL expression in a separate file. There should be 16 files in total. Place all files in **one** zip file. (There's no prescribed name for the zip file)

Submit the zip file to this Dropbox link: <https://www.dropbox.com/request/tciwzax3073dedzva1sy>.

Use the sql files accompanying this pdf file and export them to your local databases.

Database 1: Hospital Appointment System

1. **Write an SQL expression that** retrieves the names and salaries of all doctors whose specialty is 'Cardiology' and whose salary is between 100000 and 130000, ordering the results by salary in descending order.
2. **Write an SQL expression that** finds the names of all patients whose name begins with the letter 'J'.
3. **Write an SQL expression that** retrieves the doctor's name and the patient's name for every appointment, renaming the output columns to `Doctor_Name` and `Patient_Name`.
4. **Write an SQL expression that** finds the `app_id` and `app_date` of all appointments where the diagnosis has not yet been entered.
5. **Write an SQL expression that** finds the `doc_id` of all doctors who have *never* had an appointment scheduled.
6. **Write an SQL expression that** retrieves the names of all patients who have been admitted to the 'Intensive Care Unit' but have not yet been discharged.
7. **Write an SQL expression that** finds the names of doctors who have prescribed 'Lisinopril' to a patient who lives in 'Seattle'.
8. **Write an SQL expression that** lists the names of all patients who had an appointment with a doctor specializing in 'Cardiology' or 'Neurology', along with the date of the appointment and the

name of the ward they were eventually admitted to. Rename the columns to `Patient` , `Appointment_Date` , and `Ward_Name` .

Database 2: Regional Bookstore Inventory

1. **Write an SQL expression that** finds all unique countries where authors in the database are from.
2. **Write an SQL expression that** finds the `title` and `price` of all books in the 'Mystery' genre. Calculate a 10% discount on the price and output it, renaming the resulting column to `Discounted_Price` .
3. **Write an SQL expression that** finds the names of all authors who have written a book priced strictly greater than 12.00.
4. **Write an SQL expression that** generates a single list containing the `isbn` of all books that are either in the 'Horror' genre OR are part of a promotion in 'August'.
5. **Write an SQL expression that** finds the `isbn` of all books that are on promotion in *both* 'August' and 'September'.
6. **Write an SQL expression that** retrieves the titles of all books that have a current stock quantity of exactly 0 at any store branch located in 'Seattle'.
7. **Write an SQL expression that** finds the names of authors who have had at least one of their books published by a publisher based in the 'UK' and currently have a copy of that same book in stock at the 'Chicago Books' branch.
8. **Write an SQL expression that** lists the branch names, book titles, and promotional discount percentages for all 'Fantasy' books currently stocked at a branch in 'New York' that are also actively part of an 'October' promotion. Rename the output columns to `Branch` , `Book_Title` , and `October_Discount` .